

Adaptive Civic Routing Intelligence System

LIVE MVP DEPLOYED

ARCHITECTURAL DECISION RECORD (ADR) - HACKMYSURU 1.0 - ROUTING

Team Spark | ID: HM26-E9C5 | Single-Page Architectural Ledger

adaptive-civic-routing.vercel.app

Core Principle: AI assists issue categorization; PostGIS and relational rules strictly decide legal jurisdiction & department.

DECISION Q1

Deterministic PostGIS Containment vs. Probabilistic Routing

[CHOSEN DECISION]

PostgreSQL 18 + PostGIS 3.6 spatial geometry engine. Complaints store SRID

[REJECTED ALTERNATIVES]

1. Probabilistic LLM/NLP classification from complaint text (hallucinates on borderline cases)

Why Approach Was Selected:

Municipal authority (MCC vs. MUDA) is governed by statutory legal boundaries

Main Trade-off & Mitigation:

Requires maintaining verified GIS shapefiles in database. Coordinates outside municipality

DECISION Q2

Immutable Jurisdiction Versioning vs. Mutable Boundary Overwrites

[CHOSEN DECISION]

Decouple ward boundaries into explicit lifecycle states (DRAFT -> ACTIVE ->

[REJECTED ALTERNATIVES]

1. In-place polygon updates overwriting geometry coordinates directly. 2. Dynamically re-

Why Approach Was Selected:

Mysuru ward boundaries periodically delimit (e.g. MCC expanding into MUDA)

Main Trade-off & Mitigation:

Additional relational schema complexity and storage overhead to retain retired geonames

DECISION Q3

In-Database Spatial Routing & Municipal Scaling Architecture

[CHOSEN DECISION]

Execute point-in-polygon containment synchronously inside PostgreSQL at co

[REJECTED ALTERNATIVES]

1. Client-side Turf.js evaluation (leaks boundary data, vulnerable to client tampering)

Why Approach Was Selected:

In-database execution ensures ACID consistency at ticket creation (POST /api

Main Trade-off & Mitigation:

Concurrent spatial queries during civic storm events can burden database CPU/RAM